HDOC Day-44

Menu-Driven String Operations in C

Problem Statement

Write a menu-driven C program using separate functions to (1) reverse each word without changing the order of words and (2) print the initials of a name. Include an Exit option.

Algorithm

1. Display the menu in a loop and read the choice as a line.
2. Choice 1: read a sentence using fgets(); scan each whitespace-delimited word and reverse its characters in place using two indices; print the sentence.
3. Choice 2: read the name; scan characters, detect the beginning of each whitespace-delimited word, print its uppercase first letter followed by a period.
4. Choice 3: exit; otherwise show an error and repeat the menu.

Test Case Table

Test Case	Choice	Input	Expected Output
1	1	What is special?	tahW si ?laiceps
2	1	hello world	olleh dlrow
3	1	C programming is fun!	C gnimmargorp si !nuf
4	1	a bb c	a bb c
5	1	(empty sentence)	(empty output)
6	2	Alexander Grahambell	A.G.
7	2	John Ronald Reuel Tolkien	J.R.R.T.
8	2	mary jane	M.J.
9	2	(empty name)	(none)
10	9	Invalid menu choice	Invalid choice. Enter 1 to 3.
11	3	Exit	Exiting program.

Note: Reversing preserves the original letter case. Therefore, "What" becomes "tahW"; the sample in the question shows lowercase "tahw".

C Program

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>
#define MAX 500

void readLine(char text[], size_t size) {
    size_t len;
    int ch;
    if (fgets(text, (int)size, stdin) == NULL) { text[0] = '\0'; return; }
    len = strlen(text);
    if (len && text[len - 1] == '\n') text[len - 1] = '\0';
    else if (len == size - 1) while ((ch = getchar()) != '\n' && ch != EOF) { }
}

void reverseWords(void) {
```

```
    char sentence[MAX];
    size_t i = 0, start, end;
    char temp;
    printf("Enter a sentence: ");
    readLine(sentence, sizeof sentence);
    while (sentence[i] != '\0') {
        if (isspace((unsigned char)sentence[i])) { i++; continue; }
        start = i;
        while (sentence[i] != '\0' && !isspace((unsigned char)sentence[i])) i++;
        end = i;
        while (start < end && start < --end) {
            temp = sentence[start]; sentence[start] = sentence[end]; sentence[end] = temp;
            start++;
        }
    }
    printf("Reversed words: %s\n", sentence);
}

void printInitials(void) {
    char name[MAX];
    size_t i;
    int newWord = 1, found = 0;
    printf("Enter full name: ");
    readLine(name, sizeof name);
    printf("Initials: ");
    for (i = 0; name[i] != '\0'; i++) {
        unsigned char ch = (unsigned char)name[i];
        if (isspace(ch)) newWord = 1;
        else if (newWord) {
            if (isalpha(ch)) { printf("%c.", toupper(ch)); found = 1; }
            newWord = 0;
        }
    }
    if (!found) printf("(none)");
    printf("\n");
}

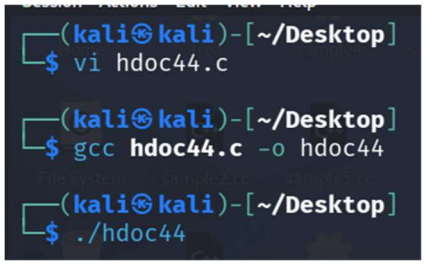
int main(void) {
    char line[64], extra;
    int choice;
    do {
        printf("\n--- STRING OPERATIONS MENU ---\n");
        printf("1. Reverse each word in a sentence\n");
        printf("2. Print initials of a name\n");
        printf("3. Exit\nEnter your choice: ");
        readLine(line, sizeof line);
        if (sscanf(line, " %d %c", &choice, &extra) != 1) {
            printf("Invalid input. Enter a number from 1 to 3.\n");
            choice = 0;
        }
        switch (choice) {
            case 1: reverseWords(); break;
            case 2: printInitials(); break;
            case 3: printf("Exiting program.\n"); break;
            case 0: break;
            default: printf("Invalid choice. Enter 1 to 3.\n");
        }
    } while (choice != 3);
    return 0;
}
```

Compilation and Execution Commands

```
gcc hdoc44.c -o hdoc44
./hdoc44
```

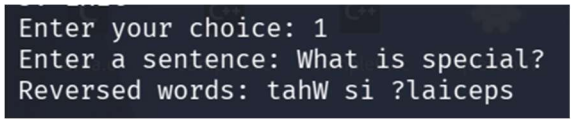
Compilation and Execution

Paste the terminal screenshot showing the compilation command and a successful run.



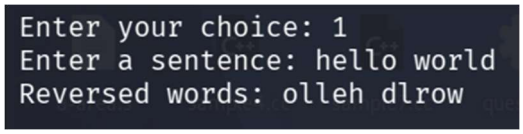
Test Case 1

Choice 1 | Input: What is special? | Expected: tahW si ?laiceps



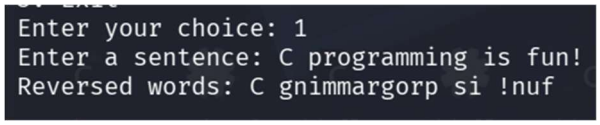
Test Case 2

Choice 1 | Input: hello world | Expected: olleh dlrow



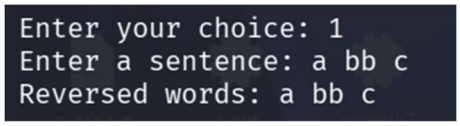
Test Case 3

Choice 1 | Input: C programming is fun! | Expected: C gnimmargorp si !nuf



Test Case 4

Choice 1 | Input: a bb c | Expected: a bb c



Test Case 5

Choice 1 | Input: (empty sentence) | Expected: (empty output)

```
Enter your choice: 1
Enter a sentence: (empty sentence)
Reversed words: ytpme( )ecnetnes
```

Test Case 6

Choice 2 | Input: Alexander Grahambell | Expected: A.G.

```
Enter your choice: 2
Enter full name: Alexander Grahambell
Initials: A.G.
```

Test Case 7

Choice 2 | Input: John Ronald Reuel Tolkien | Expected: J.R.R.T.

```
Enter your choice: 2
Enter full name: John Ronald Reuel Tolkien
Initials: J.R.R.T.
```

Test Case 8

Choice 2 | Input: mary jane | Expected: M.J.

```
Enter your choice: 2
Enter full name: mary jane
Initials: M.J.
```

Test Case 9

Choice 2 | Input: (empty name) | Expected: (none)

```
Enter your choice: 2
Enter full name:
Initials: (none)
```

Test Case 10

Choice 9 | Input: Invalid menu choice | Expected: Invalid choice. Enter 1 to 3.

```
Enter your choice: 9
Invalid choice. Enter 1 to 3.
```

Test Case 11

Choice 3 | Input: Exit | Expected: Exiting program.

```
Enter your choice: 3
Exiting program.
```